Atreyapurapu Pranathi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.61
FIITJEE JUNIOR COLLEGE	Class XII (State)	2024	98.30%
FIITJEE WORLD SCHOOL	Class X (State)	2022	9.80%

WORK EXPERIENCE

Research Intern -- Advanced Systems Laboratory (ASL), DRDO, Ministry of Defence, Government of India

May 2025 - Jul 2025

- Worked at India's premier missile structures laboratory (Dr. A.P.J. Abdul Kalam Missile Complex), fabricating advanced composite materials used in real missile and aerospace structural systems
- Executed prepreg composite layup and precision curing processes - the same manufacturing techniques behind ultra-lightweight, high-strength aerospace and defense components
- Delivered high-precision structural composite parts through hands-on fabrication, directly contributing to next-generation missile structural technology at one of India's top defense R&D institutions

Defense R&D Intern -- Composite Structures - Advanced Systems Laboratory (ASL), DRDO, Ministry of Defence, Government of India

December 2025

- Worked at India's premier missile structures laboratory (Dr. A.P.J. Abdul Kalam Missile Complex) in the domain of engineering sandwich composite structures for advanced aerospace/defense applications
- Fabricated aluminum honeycomb-core sandwich composites bonded with prepreg skins -- a construction method engineered for extreme stiffness-to-weight performance in missile and aircraft structures
- Achieved consistent, defect-free lamination and bonding quality on structural panels designed for high-performance, weight-critical defense systems

PROJECTS

Robot for Non-Destructive Testing (NDT) | Under Prof. Thulsiram G, IIT Hyderabad | July 2026- *Present*

- Designed and fabricated a remotely operated pipe-crawler robot for in-pipe non-destructive inspection, integrating drive motors, chassis, and an ultrasonic/guided-wave probe for defect detection
- Built and tuned the mechanical drivetrain (motor selection, mounting, and motion control) to enable stable crawling through constrained pipe geometries
- Developed the remote-control system for real-time robot navigation and probe positioning inside the pipe
- Implemented signal acquisition pipeline to capture wave/probe data reflecting internal pipe conditions (defects, corrosion, wall thickness variations)
- Analyzing acquired wave data to characterize internal pipe structure and detect anomalies.

Dynamic Characterization of 3D-Printed Structures using SHPB | Under Prof. Dineshkumar Harursampath, IISc Bangalore | June 2026 - Present

- Investigating the high strain-rate dynamic response of additively manufactured structures under extreme impact loading using Split Hopkinson Pressure Bar (SHPB) testing -- a technique used to simulate real-world impact, blast, and ballistic conditions
- Analyzing stress-wave propagation and dynamic stress-strain behavior to uncover how 3D-printed lattice/structural geometries absorb energy and fail under sudden high-velocity impact
- Characterizing strain-rate sensitivity, dynamic yield strength, and failure/fracture mechanisms unique to additively manufactured materials -- insights directly relevant to aerospace, defense, and crashworthy structural design
- Drafting a research paper translating this experimental data into a broader understanding of how 3D-printed structures perform under extreme dynamic loading conditions

Kinematic & Dynamic Analysis Tool for Four-Bar and Slider-Crank Mechanisms | Coursework Project | February 2026

- Developed a computational tool to analyze planar four-bar and slider-crank mechanisms, computing position, velocity, and acceleration of all links across a full input rotation cycle
- Implemented kinematic analysis using analytical methods to derive link angles, angular velocities, and angular accelerations for both mechanism types.
- Generated the coupler curve (path traced by the coupler point) for the four-bar mechanism and slider displacement/velocity/acceleration profiles for the slider-crank mechanism, visualized graphically for complete mechanism characterization
- Built using Python, enabling quick comparison of mechanism behavior across varying link lengths, crank angles, and input speeds

SKILLS

Non Destructive Testing:

Ultrasonic Testing (UT), Radiographic Testing (RT), Magnetic Particle Testing (MT).

CAE & Simulation:

LS-DYNA, Abaqus, ANSYS, MATLAB/Simulink

CAD & Design:

SolidEdge, SolidWorks, AutoCAD, Fusion360

Programming Languages:

C++, Python, C (Elementary)

Automotive:

Vehicle Dynamics, Structural Analysis, Materials Selection, Mechatronics & Automation

RELEVANT COURSES

Relevant Courses: Engineering Mechanics, Strength of Materials, Thermodynamics, Fluid Mechanics, Heat and Mass Transfer, Introduction to Aero Vehicles, Manufacturing Science, Fundamentals of Physical Metallurgy, Kinematics and Dynamics of Machinery

POSITIONS OF RESPONSIBILITY

Inbound Buddy International Cell

- Coordinated logistics and onboarding for international delegates during campus visits.
- Managed guest communication and collaborated with teams to ensure seamless program execution.
- Facilitated cross-cultural engagement through efficient stakeholder and event coordination.

PR Head, International Cell

- Worked closely with the **Japan Desk to coordinate India-Japan student exchange programs**, managing communication and coordination for both **outbound students from India and incoming students from Japan** throughout the exchange process.
- Led branding, outreach, and external communications for the International Relations Cell, strengthening its visibility and engagement across campus.
- Managed collaborations with student bodies and external partners, while planning promotional campaigns to increase participation in international programs and events.

Internship Coordinator, Office of Career Services

- Verified student CVs to ensure compliance with OCS guidelines and improve application quality.
- Coordinated communication and logistics between recruiters, students, and the OCS team.
- Assisted in organizing placement drives and supporting the smooth execution of recruitment activities